

YEAR 7 MATHEMATICS

Chapter 2

Angles

MIXED PRACTICE WORKSHEETS

2

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

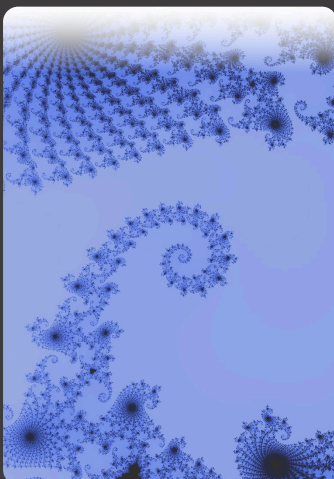
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
2.01	Naming angles	3		☐
2.02	Measuring and drawing angles	5		☐
2.03	Classifying angles	7		☐
2.04	Complementary and supplementary angles	9		☐
2.05	Angles at a point and vertically opposite angles	11		☐
2.06	Parallel and perpendicular lines	13		☐
2.07	Corresponding angles	15		☐
2.08	Alternate angles	17		☐
2.09	Co-interior angles	19		☐
2.10	Finding unknown angles on parallel lines	21		☐
2.11	Proving lines are parallel	23		☐

In this chapter you will

- measure and draw angles with a protractor
- classify angles by size
- find complementary, supplementary and vertically opposite angles
- construct parallel and perpendicular lines
- find corresponding, alternate and co-interior angles on parallel lines
- prove that two lines are parallel



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is Seahorse Valley, the curling tails between the main body and the round bulb on its left. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

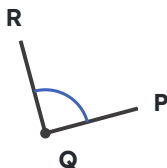
WE DO

Watch your teacher.
Copy every step.

KEY IDEA

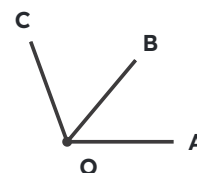
Name an angle with three letters, with the vertex (the corner) in the middle: $\angle ABC$ has vertex B. Adjacent angles share a vertex and an arm.

E1 Name the marked angle.



E2 Name the vertex and the arms of $\angle XYZ$.

E3 Name the three angles at O.

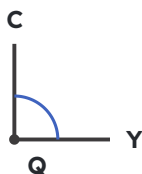


Easy

Questions 1–7

1 Which letter is the vertex of $\angle AUR$?

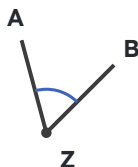
2 Name the marked angle with three letters.



3 Write $\angle WPV$ another way.

4 Which letter is the vertex of $\angle DER$?

5 Name the marked angle with three letters.



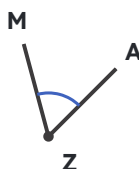
6 Write $\angle GPL$ another way.

7 Which letter is the vertex of $\angle WVC$?

Easy

Questions 8–14

8 Name the marked angle with three letters.



9 Write $\angle XKN$ another way.

10 Which letter is the vertex of $\angle WMP$?

11 What is the opposite of -26 ?

from 1.01

12 Name the vertex and the two arms of $\angle DSU$.

vertex: _____

arms: _____

13 Name the marked angle with three letters.



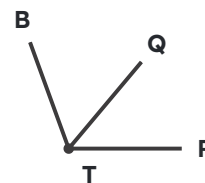
14 Write $\angle XCE$ another way.

Medium

Questions 15–20

15 Name the vertex and the two arms of $\angle CHB$.

16 Name the three angles at T.

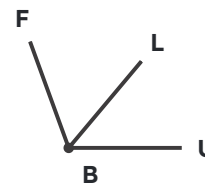


17 Start at 6. Move 7 left. Where do you land?

from 1.02

18 $\angle HCP$ and $\angle PCN$ are adjacent. Name the arm they share.

19 Name a pair of adjacent angles in the diagram.



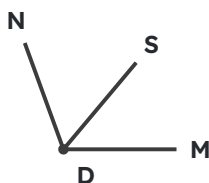
20 Use a calculator: $(-22)^2$

from 1.09

Medium

Questions 21–28

- 21 Name the three angles at D.

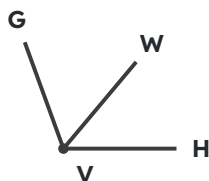


- 22 $\angle HRN$ and $\angle NRZ$ are adjacent. Name the arm they share.

- 23 $-2 \times (-8)$

from 1.06

- 24 Name a pair of adjacent angles in the diagram.



- 25 $12 + (-17)$

from 1.04

- 26 Name the vertex and the two arms of $\angle SNX$.

- 27 $\angle CUZ = 42^\circ$ and $\angle ZUX = 52^\circ$ are adjacent. They do not overlap. Find $\angle CUX$.

- 28 $9 - (-4)$

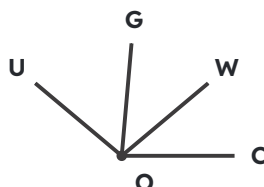
from 1.05

Challenging

Questions 29–35

- 29 $\angle DQR = 46^\circ$ and $\angle RQV = 49^\circ$ are adjacent. They do not overlap. Find $\angle DQV$.

- 30 How many different angles are at O? Name them.



- 31 Which is larger: -16 or 5 ?

from 1.03

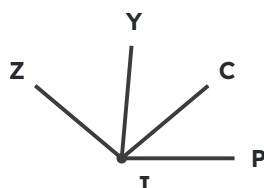
- 32 Are $\angle SNF$ and $\angle BEF$ adjacent? Explain.

- 33 $(-3)^2 - 23$

from 1.08

- 34 $\angle HZQ = 21^\circ$ and $\angle QZA = 26^\circ$ are adjacent. They do not overlap. Find $\angle HZA$.

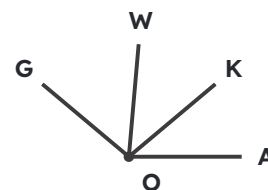
- 35 How many different angles are at J? Name them.



Challenging

Questions 36–41

- 36 How many different angles are at O? Name them.



- 37 Are $\angle SHR$ and $\angle FQC$ adjacent? Explain.

- 38 How far is it from -3 to 3 ?

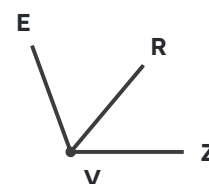
from 1.02

- 39 $\angle RWD = 60^\circ$ and $\angle DWP = 27^\circ$ are adjacent. They do not overlap. Find $\angle RWP$.

- 40 Use a calculator:
 $(-198 + 30) \times (-13)$

from 1.09

- 41 Name a pair of adjacent angles in the diagram.



NAME _____ CLASS _____ DATE _____

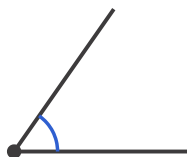
WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Put the centre of the protractor on the vertex. Line up 0° with one arm, then count up from 0° on that scale. For a reflex angle, measure the angle inside and take it from 360° .

E1 Measure the angle.



E2 Draw an angle of 130° on the ray.

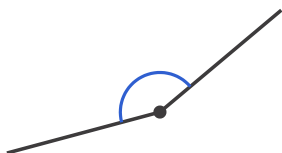


E3 How do you draw a reflex angle of 250° ?

Easy

Questions 1–6

1 Measure the angle.

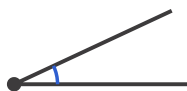


2 To draw a reflex angle of 200° , what angle do you draw first?

$\underline{\quad}^\circ - \underline{\quad}^\circ$
 $= \underline{\quad}^\circ$

3 Is 165° more or less than a right angle?

4 Measure the angle.



5 To draw a reflex angle of 225° , what angle do you draw first?

6 Is 50° more or less than a right angle?

Easy

Questions 7–12

7 To draw a reflex angle of 290° , what angle do you draw first?

$\underline{\quad}^\circ - \underline{\quad}^\circ$
 $= \underline{\quad}^\circ$

8 Is 145° more or less than a right angle?

9 Measure the angle.



10 Aisha has \$61 in the bank and spends \$69. What is the balance?

from 1.10

11 Estimate the angle. Then measure it.

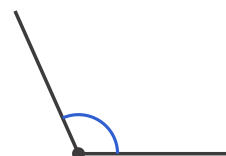


12 To draw a reflex angle of 320° , what angle do you draw first?

Medium

Questions 13–16

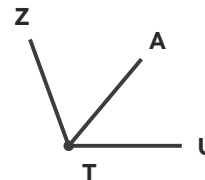
13 Estimate the angle. Then measure it.



14 Draw an angle of 45° on the ray.

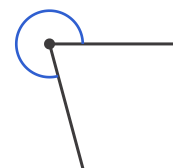


15 Name a pair of adjacent angles in the diagram.



from 2.01

16 Measure the reflex angle (the one marked outside).



Medium

Questions 17–21

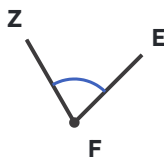
- 17 Draw an angle of 135° on the ray.



- 18 Measure the reflex angle (the one marked outside).

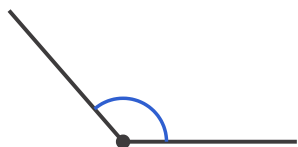


- 19 Name the marked angle with three letters.

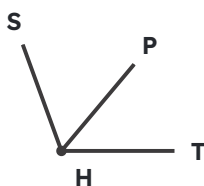


from 2.01

- 20 Estimate the angle. Then measure it.



- 21 Name the three angles at H.



from 2.01

Challenging

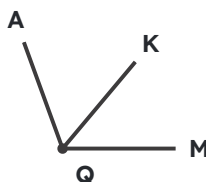
Questions 22–28

- 22 Draw a reflex angle of 230° on the ray.



- 23 An angle is 70° less than a revolution. Find its size and classify it.

- 24 Name a pair of adjacent angles in the diagram.



from 2.01

- 25 The minute hand turns from 5 to 11 on a clock. Through what angle does it turn?

- 26 Which is larger: 4 or -9 ?

from 1.03

- 27 Ruby measured an angle as 155° , but it is clearly acute. What mistake was made? What is the angle?

- 28 Draw a reflex angle of 220° on the ray.



Challenging

Questions 29–35

- 29 An angle is 30° less than a revolution. Find its size and classify it.

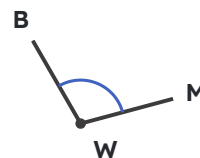
- 30 The minute hand turns from 1 to 3 on a clock. Through what angle does it turn?

- 31 Which integer is halfway between -10 and 0 ?

from 1.02

- 32 Jack measured an angle as 115° , but it is clearly acute. What mistake was made? What is the angle?

- 33 Name the marked angle with three letters.



from 2.01

- 34 Draw a reflex angle of 205° on the ray.



- 35 Measure the reflex angle (the one marked outside).



NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Acute: less than 90° . Right: 90° . Obtuse: between 90° and 180° . Straight: 180° . Reflex: between 180° and 360° . Revolution: 360° .

E1 Classify 135° .

E2 Classify 270° .

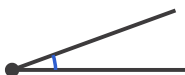
E3 Classify the angle between the clock hands at 4:00.

Easy

Questions 1–8

1 Classify an angle of 270° .

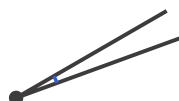
2 Classify the angle.



3 Write the size of any obtuse angle.

4 Classify an angle of 350° .

5 Classify the angle.



6 Write the size of any acute angle.

7 Classify an angle of 180° .

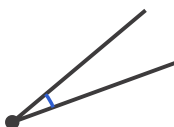
8 Classify the angle.



Easy

Questions 9–16

9 Classify the angle.



10 Write the size of any reflex angle.

11 Classify an angle of 30° .

12 $5 + (-6)$

from 1.04

13 Add the angles, then classify the total: $15^\circ + 45^\circ$

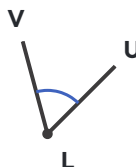
____ $^\circ$ + ____ $^\circ$ = ____ $^\circ$

14 Classify the angle.



15 Classify an angle of 360° .

16 Name the marked angle with three letters.



from 2.01

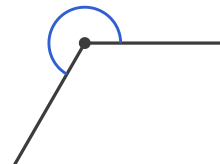
Medium

Questions 17–23

17 Add the angles, then classify the total: $30^\circ + 75^\circ$

18 Find the angle between the clock hands at 2:00. Classify it.

19 Measure the reflex angle (the one marked outside).



from 2.02

20 Find a straight angle plus 85° . Classify it.

21 A reflex angle is 210° . Find the other angle at the vertex and classify it.

22 Which is larger: 1 or -5 ?

from 1.03

23 Add the angles, then classify the total: $15^\circ + 10^\circ$

Medium

Questions 24–32

- 24 Find the angle between the clock hands at 8:00. Classify it.

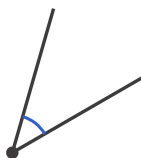
- 25 Find a right angle plus 60° . Classify it.

- 26 Write $\angle VPF$ another way.

from 2.01

- 27 A reflex angle is 205° . Find the other angle at the vertex and classify it.

- 28 Estimate the angle. Then measure it.



from 2.02

- 29 Add the angles, then classify the total: $35^\circ + 165^\circ$

- 30 Can two obtuse angles add to a reflex angle? Explain.

- 31 Find the average of these temperatures: -12°C , -8°C , -1°C , 1°C

from 1.07

- 32 Find the angle between the clock hands at 4:00. Classify it.

Challenging

Questions 33–40

- 33 Can an obtuse angle and an acute angle add to a right angle? Explain.

- 34 An angle is the size of five angles of 45° together. Find its size and classify it.

- 35 Draw an angle of 160° on the ray.



from 2.02

- 36 A revolution is cut into 6 equal angles. Find one angle and classify it.

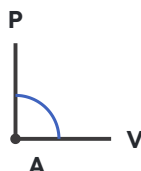
- 37 $6 \times 6 - 6$

from 1.08

- 38 The hands of a clock show 3:00. Find the reflex angle between them.

- 39 Can two acute angles add to a reflex angle? Explain.

- 40 Name the marked angle with three letters.



from 2.01

Challenging

Questions 41–49

- 41 An angle is the size of one and a half right angles. Find its size and classify it.

- 42 A revolution is cut into 4 equal angles. Find one angle and classify it.

- 43 Use a calculator: $(-22)^2$

from 1.09

- 44 The hands of a clock show 1:00. Find the reflex angle between them.

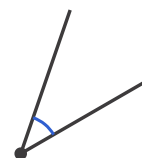
- 45 Write $\angle WBS$ another way.

from 2.01

- 46 A reflex angle is 310° . Find the other angle at the vertex and classify it.

- 47 An angle is the size of three angles of 60° together. Find its size and classify it.

- 48 Estimate the angle. Then measure it.



from 2.02

- 49 A revolution is cut into 8 equal angles. Find one angle and classify it.

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Complementary angles add to 90° . Supplementary angles add to 180° . To find the missing angle, take it away from 90° or from 180° .

E1 Find the complement of 35° .

E2 Find the supplement of 112° .

E3 Find x .



Easy

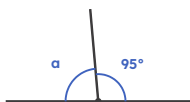
Questions 1–8

1 Find the complement of 66° .

2 Find the supplement of 70° .

3 Are 66° and 34° complementary?

4 Find a .



5 Find the complement of 63° .

6 Find the supplement of 118° .

7 Are 68° and 32° complementary?

8 Find a .



Easy

Questions 9–16

9 Find the supplement of 109° .

10 Are 80° and 20° complementary?

11 Find a .



12 Leo has \$47 in the bank and spends \$105. What is the balance?

from 1.10

13 Find the complement of 36° .

14 m° and 73° are complementary. Find m .

$$m + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$m = \underline{\hspace{1cm}}$$

15 Find the supplement of 122° .

16 To draw a reflex angle of 325° , what angle do you draw first?

from 2.02

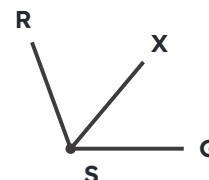
Medium

Questions 17–22

17 y° and 50° are complementary. Find y .

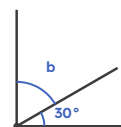
18 x° and 79° are supplementary. Find x .

19 Name the three angles at S.



from 2.01

20 Find b .



21 Are 149° and 31° supplementary? Explain.

22 Which is larger: 3 or -18 ?

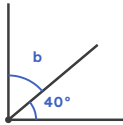
from 1.03

Medium

Questions 23–30

- 23 x° and 51° are supplementary. Find x .

- 24 Find b .

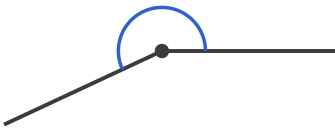


- 25 Write the size of any reflex angle.

from 2.03

- 26 Are 157° and 23° supplementary? Explain.

- 27 Measure the reflex angle (the one marked outside).



from 2.02

- 28 y° and 56° are complementary. Find y .

- 29 Find x .



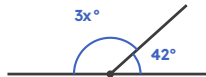
- 30 Find the average of these temperatures: 4°C , -5°C , 1°C

from 1.07

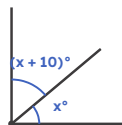
Challenging

Questions 31–37

- 31 Find x .



- 32 Find x .



- 33 A reflex angle is 320° . Find the other angle at the vertex and classify it.

from 2.03

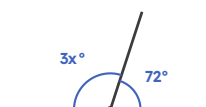
- 34 An angle is twice its complement. Find the angle.

- 35 What is the opposite of -21 ?

from 1.01

- 36 Find the supplement of the complement of 25° .

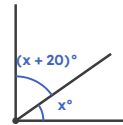
- 37 Find x .



Challenging

Questions 38–44

- 38 Find x .



- 39 An angle is three times its supplement. Find the angle.

- 40 $17 + (-22)$

from 1.04

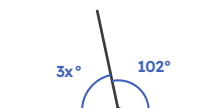
- 41 Find the supplement of the complement of 67° .

- 42 Classify the angle.



from 2.03

- 43 Find x .



- 44 Are 132° and 48° supplementary? Explain.

NAME _____ CLASS _____ DATE _____

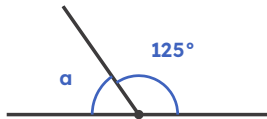
WE DO

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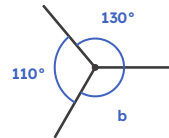
KEY IDEA

Angles on a straight line add to 180° . Angles at a point add to 360° . Vertically opposite angles are equal.

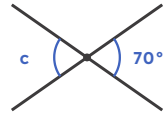
E1 Find a .



E2 Find b .



E3 Find c . Give a reason.



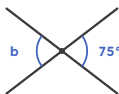
Easy

Questions 1–6

1 Find a .



2 Find b .

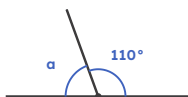


3 Angles at a point are 85° , 95° and c° . Find c .

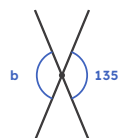
$c = ______ - ______ - ______$

$c = ______$

4 Find a .



5 Find b .

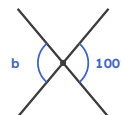


6 Angles at a point are 65° , 40° and c° . Find c .

Easy

Questions 7–12

7 Find b .

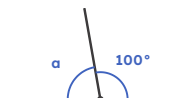


8 Angles at a point are 45° , 50° and c° . Find c .

$c = ______ - ______ - ______$

$c = ______$

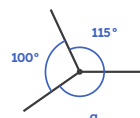
9 Find a .



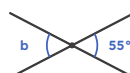
10 What is the opposite of -6 ?

from 1.01

11 Find a .



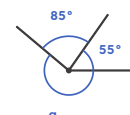
12 Find b .



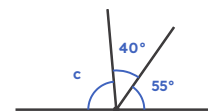
Medium

Questions 13–16

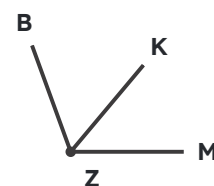
13 Find a .



14 Find c .

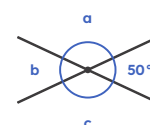


15 Name the three angles at Z .



from 2.01

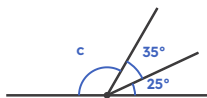
16 Find a , b and c .



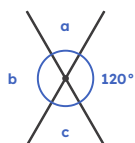
Medium

Questions 17–24

- 17 Find c .



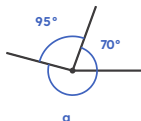
- 18 Find a , b and c .



- 19 Use a calculator: -51×32

from 1.09

- 20 Find a .



- 21 Write $\angle NGB$ another way.

from 2.01

- 22 Angles at a point are x° , $2x^\circ$ and $3x^\circ$. Find x .

- 23 Find c .



- 24 Are 50° and 40° complementary?

from 2.04

Challenging

Questions 25–32

- 25 Angles at a point are $2x^\circ$, $3x^\circ$ and $4x^\circ$. Find x .

- 26 Vertically opposite angles are $(2x + 15)^\circ$ and 83° . Find x .

- 27 The temperature went from -1°C to 7°C . How much did it rise?

from 1.05

- 28 On a straight line, the angles are 132° , x° and $3x^\circ$. Find x .

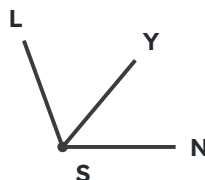
- 29 Is 95° more or less than a right angle?

from 2.02

- 30 Chloe says the angle next to a 50° angle on a straight line is also 50° . Is that right? Explain.

- 31 Angles at a point are x° , $3x^\circ$ and $5x^\circ$. Find x .

- 32 Name a pair of adjacent angles in the diagram.



from 2.01

Challenging

Questions 33–40

- 33 Vertically opposite angles are $(2x + 15)^\circ$ and 53° . Find x .

- 34 On a straight line, the angles are 100° , x° and $3x^\circ$. Find x .

- 35 $-6 \times (-8)$

from 1.06

- 36 Jack says the angle next to a 40° angle on a straight line is also 40° . Is that right? Explain.

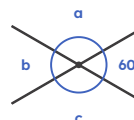
- 37 Name the marked angle with three letters.



from 2.01

- 38 Angles at a point are x° , x° and $2x^\circ$. Find x .

- 39 Find a , b and c .



- 40 Classify an angle of 90° .

from 2.03

NAME _____ CLASS _____ DATE _____

WE DO

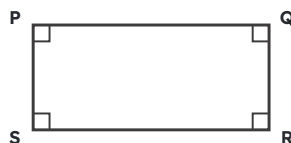
KEY IDEA

Parallel lines ($\parallel$) never meet: they stay the same distance apart. Perpendicular lines ($\perp$) meet at a right angle.

Watch your teacher.
Copy every step.

E1 Write with a symbol: AB is perpendicular to CD.

E2 Name the parallel sides of rectangle PQRS.



E3 Construct a line perpendicular to AB through P, a point on AB.

Easy

Questions 1–9

1 Write with a symbol: HX is perpendicular to WY.

2 Are the lines in train tracks parallel, perpendicular or neither?

3 Write in words: $LC \parallel TV$

4 Write with a symbol: HB is perpendicular to DZ.

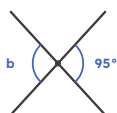
5 Are the lines in the letter T parallel, perpendicular or neither?

6 Write in words: $SE \perp WC$

7 Write with a symbol: CR is perpendicular to WD.

8 Are the lines in the letter X parallel, perpendicular or neither?

9 Find b .



from 2.05

Easy

Questions 10–17

10 Are the lines in an equals sign parallel, perpendicular or neither?

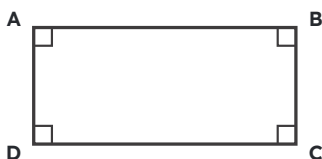
11 Write in words: $TL \parallel XN$

12 Write with a symbol: PU is perpendicular to GD.

13 Use a calculator: -15×31

from 1.09

14 Name the parallel sides and the perpendicular sides of rectangle ABCD.



$\parallel$: _____

$\perp$: _____

15 Are the lines in a plus sign parallel, perpendicular or neither?

16 Write in words: $CD \parallel HK$

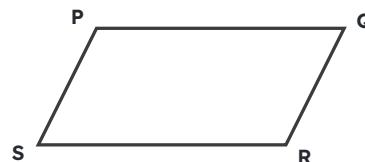
17 Find the supplement of 38° .

from 2.04

Medium

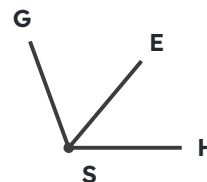
Questions 18–21

18 Name the parallel sides and the perpendicular sides of parallelogram PQRS.



19 $GW \perp PC$. What size is the angle where they meet?

20 Name the three angles at S.



from 2.01

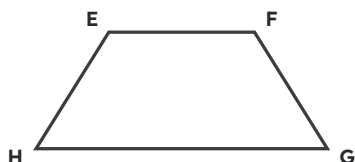
21 Draw two lines that are perpendicular. Mark them with the right symbol.

Medium

Questions 22–28

- 22 $DY \perp HU$. What size is the angle where they meet?

- 23 Name the parallel sides and the perpendicular sides of trapezium EFGH.



- 24 $2 \times (-4) - 6$

from 1.08

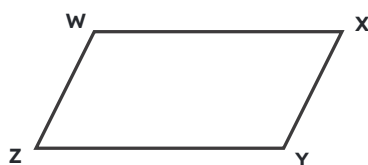
- 25 Write the steps to construct a line perpendicular to PQ through the point V on PQ.

- 26 To draw a reflex angle of 230° , what angle do you draw first?

from 2.02

- 27 $TN \perp LA$. What size is the angle where they meet?

- 28 Name the parallel sides and the perpendicular sides of parallelogram WXYZ.



Challenging

Questions 29–35

- 29 Construct a line perpendicular to the ray at its end point. Use compasses.



- 30 Write the steps to construct a line parallel to PQ through the point V, which is not on PQ.

- 31 Find the average of these temperatures: -3°C , -7°C , 4°C , 6°C

from 1.07

- 32 $AB \parallel CD$ and $CD \perp EF$. What can you say about AB and EF?

- 33 Write the size of any reflex angle.

from 2.03

- 34 Write the steps to construct the perpendicular bisector of the interval PQ.

- 35 In rectangle WXYZ, which sides are perpendicular to WX? Which side is parallel to WX?

Challenging

Questions 36–41

- 36 $AB \perp CD$ and $CD \perp EF$. What can you say about AB and EF?

- 37 In rectangle ABCD, which sides are perpendicular to AB? Which side is parallel to AB?

- 38 $-5 - (-5)$

from 1.05

- 39 $AB \parallel CD$ and $CD \parallel EF$. What can you say about AB and EF?

- 40 Is 60° more or less than a right angle?

from 2.02

- 41 In rectangle PQRS, which sides are perpendicular to PQ? Which side is parallel to PQ?

NAME _____ CLASS _____ DATE _____

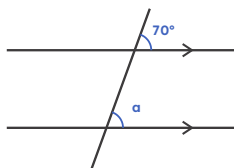
WE DO

KEY IDEA

When a line crosses two parallel lines, corresponding angles are equal. They are in the same position at each crossing: look for an F shape.

Watch your teacher.
Copy every step.

- E1** The lines are parallel. Find a .



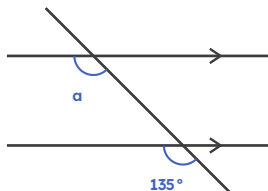
- E2** Find b , the angle next to a on the line.

- E3** Corresponding angles are $(2x + 10)^\circ$ and 70° . Find x .

Easy

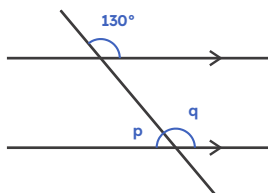
Questions 1–5

- 1** The lines are parallel. Find a .

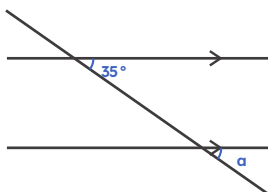


- 2** The lines are parallel. An angle is 140° . What is its corresponding angle?

- 3** The lines are parallel. Which angle, p or q , corresponds to the 130° angle?



- 4** The lines are parallel. Find a .



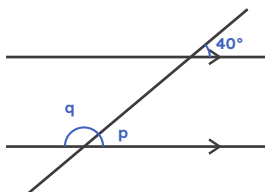
- 5** The lines are parallel. An angle is 24° . What is its corresponding angle?

Easy

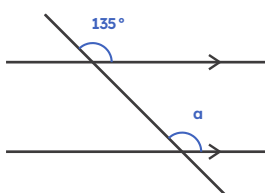
Questions 6–11

- 6** The lines are parallel. An angle is 48° . What is its corresponding angle?

- 7** The lines are parallel. Which angle, p or q , corresponds to the 40° angle?



- 8** The lines are parallel. Find a .



- 9** Use a calculator: -29×39

from 1.09

- 10** a is corresponding to 140° , and b is next to a on a straight line. Find a and b .

$a = \underline{\hspace{2cm}}$ (corresponding)

$b = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

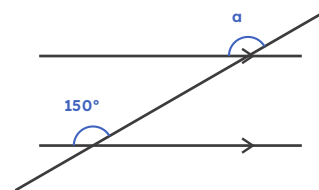
- 11** The lines are parallel. An angle is 44° . What is its corresponding angle?

Medium

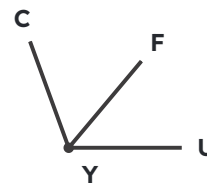
Questions 12–16

- 12** a is corresponding to 40° , and b is next to a on a straight line. Find a and b .

- 13** Find a . Give a reason.



- 14** Name the three angles at Y .



from 2.01

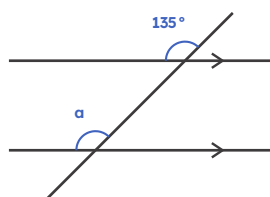
- 15** Corresponding angles are $(x + 5)^\circ$ and 25° . Find x .

- 16** a is corresponding to 145° , and b is next to a on a straight line. Find a and b .

Medium

Questions 17–23

- 17 Find a . Give a reason.



- 18 Corresponding angles are $(x + 5)^\circ$ and 10° . Find x .

- 19 $4 \times (-6) - 10$

from 1.08

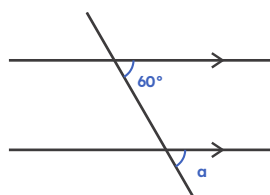
- 20 a is corresponding to 90° , and b is next to a on a straight line. Find a and b .

- 21 To draw a reflex angle of 350° , what angle do you draw first?

from 2.02

- 22 Corresponding angles are $(4x + 5)^\circ$ and 49° . Find x .

- 23 Find a . Give a reason.

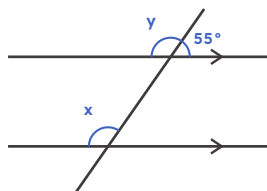


Challenging

Questions 24–30

- 24 Corresponding angles are $(4x + 5)^\circ$ and 73° . Find x .

- 25 Find y , then x . Give reasons.



- 26 Which integer is halfway between -9 and 3 ?

from 1.02

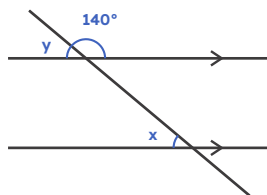
- 27 Noah says corresponding angles always add to 180° . Is that right? Explain.

- 28 Are 55° and 35° complementary?

from 2.04

- 29 Corresponding angles are $(2x + 25)^\circ$ and 61° . Find x .

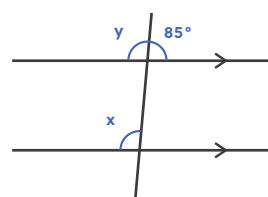
- 30 Find y , then x . Give reasons.



Challenging

Questions 31–38

- 31 Find y , then x . Give reasons.



- 32 Corresponding angles are $(3x + 5)^\circ$ and 74° . Find x .

- 33 Ali has \$75 in the bank and spends \$86. What is the balance?

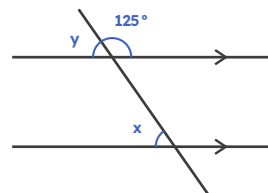
from 1.10

- 34 Corresponding angles are $(x + 5)^\circ$ and 19° . Find x .

- 35 Write $\angle HXV$ another way.

from 2.01

- 36 Find y , then x . Give reasons.



- 37 Corresponding angles are $(4x + 15)^\circ$ and 39° . Find x .

- 38 Write with a symbol: ZK is parallel to HX.

from 2.06

NAME _____ CLASS _____ DATE _____

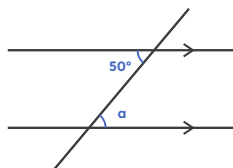
WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Alternate angles are between the parallel lines, on opposite sides of the line that crosses them. They are equal: look for a Z shape.

- E1** The lines are parallel. Find a .



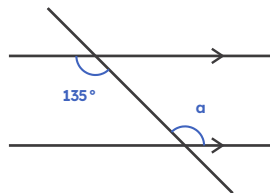
- E2** a is alternate to 115° . Find a , then b next to it on the line.

- E3** Alternate angles are $(3x - 15)^\circ$ and 60° . Find x .

Easy

Questions 1–6

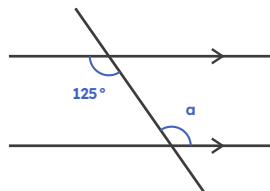
- 1** The lines are parallel. Find a .



- 2** The lines are parallel. What is the alternate angle to 106° ?

- 3** Are alternate angles between the parallel lines or outside them?

- 4** The lines are parallel. Find a .



- 5** The lines are parallel. What is the alternate angle to 49° ?

- 6** What letter shape do alternate angles make?

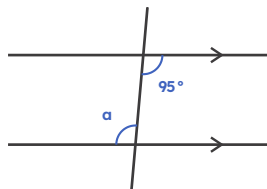
Easy

Questions 7–12

- 7** The lines are parallel. What is the alternate angle to 32° ?

- 8** Are alternate angles on the same side of the crossing line, or on opposite sides?

- 9** The lines are parallel. Find a .



- 10** Lena has \$70 in the bank and spends \$79. What is the balance?

from 1.10

- 11** a is alternate to 155° , and b is next to a on the line. Find a and b .

$a = \underline{\hspace{2cm}}$ (alternate)

$b = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

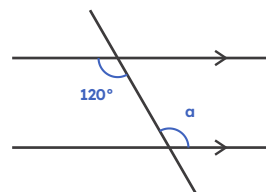
- 12** The lines are parallel. What is the alternate angle to 88° ?

Medium

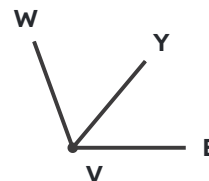
Questions 13–17

- 13** a is alternate to 125° , and b is next to a on the line. Find a and b .

- 14** Find a . Give a reason.



- 15** Name the three angles at V.



from 2.01

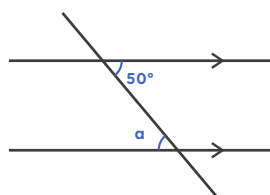
- 16** Alternate angles are $(x - 15)^\circ$ and 12° . Find x .

- 17** a is alternate to 30° , and b is next to a on the line. Find a and b .

Medium

Questions 18–24

- 18 Find a . Give a reason.



- 19 Alternate angles are $(x - 25)^\circ$ and 0° . Find x .

- 20 What is the opposite of -30 ?

from 1.01

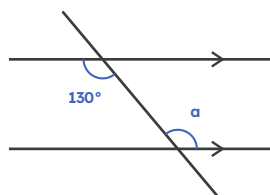
- 21 a is alternate to 140° , and b is next to a on the line. Find a and b .

- 22 To draw a reflex angle of 275° , what angle do you draw first?

from 2.02

- 23 Alternate angles are $(3x - 15)^\circ$ and 30° . Find x .

- 24 Find a . Give a reason.

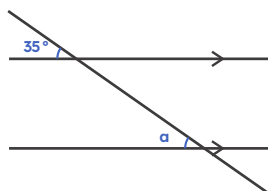


Challenging

Questions 25–30

- 25 Alternate angles are $(2x - 25)^\circ$ and 29° . Find x .

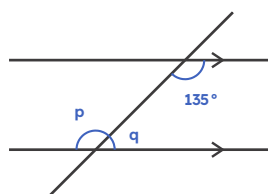
- 26 Find a . Is it a corresponding or an alternate angle?



- 27 The temperature went from -3°C to 2°C . How much did it rise?

from 1.05

- 28 Find p , then q . Give reasons.



- 29 Angles at a point are 65° , 40° and c° . Find c .

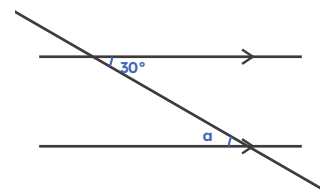
from 2.05

- 30 Alternate angles are $(2x - 5)^\circ$ and 35° . Find x .

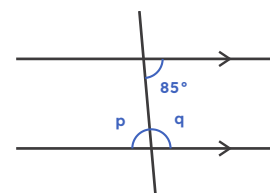
Challenging

Questions 31–37

- 31 Find a . Is it a corresponding or an alternate angle?



- 32 Find p , then q . Give reasons.



- 33 $1 + (-6)$

from 1.04

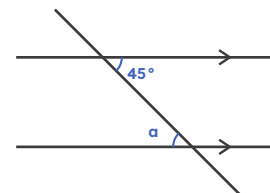
- 34 Alternate angles are $(4x - 5)^\circ$ and 107° . Find x .

- 35 Find the supplement of 142° .

from 2.04

- 36 Alternate angles are $(x - 30)^\circ$ and 7° . Find x .

- 37 Find a . Is it a corresponding or an alternate angle?



NAME _____ CLASS _____ DATE _____

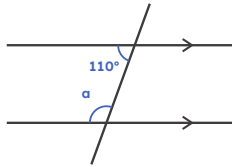
WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Co-interior angles are between the parallel lines, on the same side of the line that crosses them. They add to 180° : look for a C shape.

- E1** The lines are parallel. Find a .



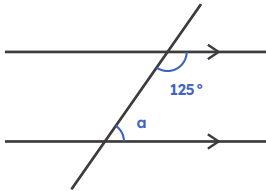
- E2** a is co-interior to 65° . Find a .

- E3** Co-interior angles are y° and $2y^\circ$. Find y .

Easy

Questions 1–6

- 1** The lines are parallel. Find a .

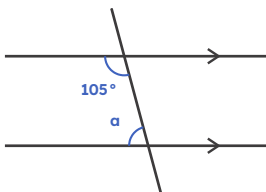


- 2** The lines are parallel. What is the co-interior angle to 83° ?

_____° - _____°
= _____°

- 3** Do co-interior angles add to 90° , add to 180° , or are they equal?

- 4** The lines are parallel. Find a .



- 5** The lines are parallel. What is the co-interior angle to 124° ?

- 6** Are co-interior angles on the same side of the crossing line?

Easy

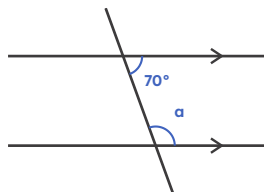
Questions 7–12

- 7** The lines are parallel. What is the co-interior angle to 148° ?

_____° - _____°
= _____°

- 8** What letter shape do co-interior angles make?

- 9** The lines are parallel. Find a .



- 10** Start at 1. Move 8 left. Where do you land?

from 1.02

- 11** a is co-interior to 128° . Write an equation and find a .

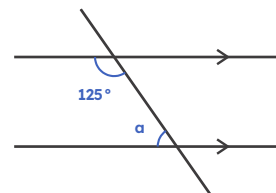
- 12** The lines are parallel. What is the co-interior angle to 34° ?

Medium

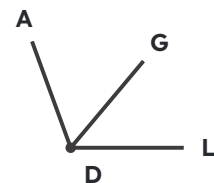
Questions 13–17

- 13** a is co-interior to 158° . Write an equation and find a .

- 14** Find a . Give a reason.



- 15** Name the three angles at D.



from 2.01

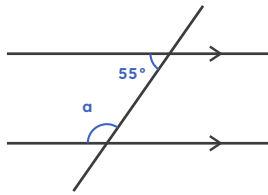
- 16** Can 102° and 78° be co-interior angles on parallel lines? Explain.

- 17** a is co-interior to 47° . Write an equation and find a .

Medium

Questions 18–25

- 18 Find a . Give a reason.



- 19 Can 32° and 138° be co-interior angles on parallel lines? Explain.

- 20 Use a calculator: -46×33

from 1.09

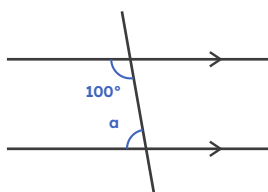
- 21 a is co-interior to 39° . Write an equation and find a .

- 22 To draw a reflex angle of 230° , what angle do you draw first?

from 2.02

- 23 Co-interior angles are y° and $5y^\circ$. Find y and both angles.

- 24 Find a . Give a reason.



- 25 Are alternate angles on the same side of the crossing line, or on opposite sides?

from 2.08

Challenging

Questions 26–33

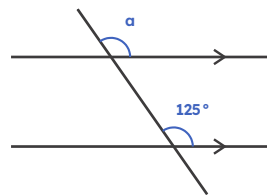
- 26 Co-interior angles are y° and $3y^\circ$. Find y and both angles.

- 27 Co-interior angles are $(2x + 20)^\circ$ and 118° . Find x .

- 28 $2 - (4 - 5)$

from 1.08

- 29 Find a . Name the angle pair.



- 30 Write in words: $SZ \parallel NR$

from 2.06

- 31 Co-interior angles are y° and $4y^\circ$. Find y and both angles.

- 32 Co-interior angles are $(2x + 30)^\circ$ and 106° . Find x .

- 33 Find the angle between the clock hands at 1:00. Classify it.

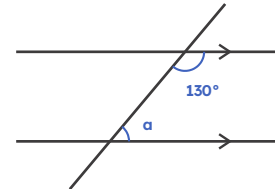
from 2.03

Challenging

Questions 34–41

- 34 Co-interior angles are $(2x + 20)^\circ$ and 126° . Find x .

- 35 Find a . Name the angle pair.



- 36 $-5 \times (-5)$

from 1.06

- 37 Co-interior angles are y° and $8y^\circ$. Find y and both angles.

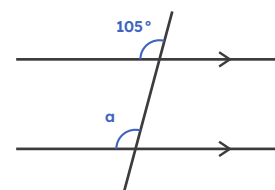
- 38 Find the supplement of 136° .

from 2.04

- 39 Can 45° and 125° be co-interior angles on parallel lines? Explain.

- 40 Co-interior angles are $(2x + 10)^\circ$ and 148° . Find x .

- 41 The lines are parallel. Find a .



from 2.07

NAME _____ CLASS _____ DATE _____

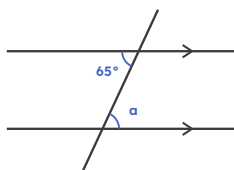
WE DO

KEY IDEA

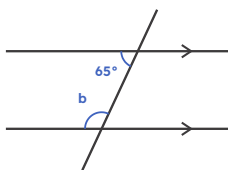
To find an unknown angle on parallel lines: spot the pair (F corresponding, Z alternate, C co-interior), use the rule, and give the reason.

Watch your teacher.
Copy every step.

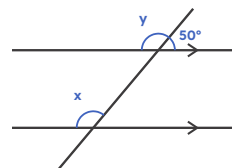
E1 Which rule? Find a .



E2 Find b . Give a reason.



E3 Find y , then x . Give reasons.

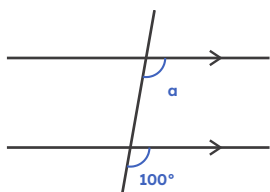


Easy

Questions 1–7

1 Which angle pair is an F shape?

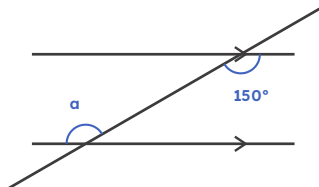
2 The lines are parallel. Find a .



3 Complete the rule: corresponding angles on parallel lines ...

4 Which angle pair is a Z shape?

5 The lines are parallel. Find a .



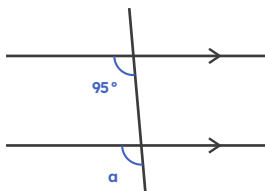
6 Complete the rule: alternate angles on parallel lines ...

7 Which angle pair is a C shape?

Easy

Questions 8–12

8 The lines are parallel. Find a .



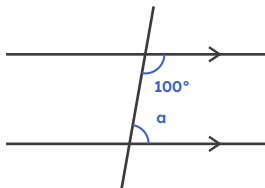
9 Complete the rule: co-interior angles on parallel lines ...

10 Which angle pair is angles in the same position at each crossing?

11 $7 - (-5)$

from 1.05

12 Find a . Write the rule you used.



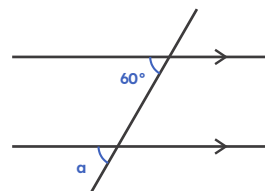
$a = \underline{\quad} - \underline{\quad} = \underline{\quad}$

co-interior angles add to $\underline{\quad}^\circ$

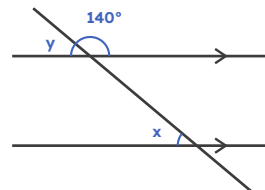
Medium

Questions 13–16

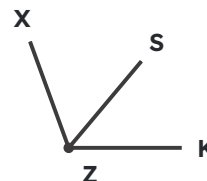
13 Find a . Write the rule you used.



14 Find y , then x . Give reasons.



15 Name the three angles at Z.



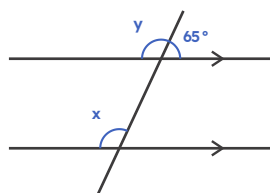
from 2.01

16 m and 80° are co-interior angles on parallel lines. Find m .

Medium

Questions 17–22

- 17 Find y , then x . Give reasons.

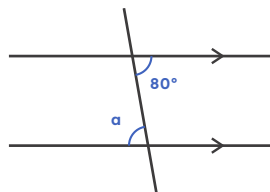


- 18 m and 53° are alternate angles on parallel lines. Find m .

- 19 Zac has \$44 in the bank and spends \$60. What is the balance?

from 1.10

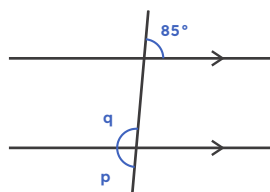
- 20 Find a . Write the rule you used.



- 21 To draw a reflex angle of 240° , what angle do you draw first?

from 2.02

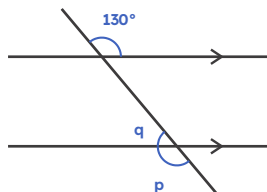
- 22 Find p and q . Give a reason for each.



Challenging

Questions 23–27

- 23 Find p and q . Give a reason for each.



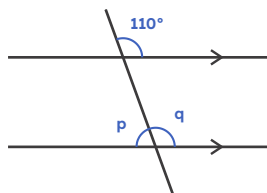
- 24 Alternate angles on parallel lines are $(3x + 15)^\circ$ and 75° . Find x .

- 25 Find the average of these temperatures: -8°C , 5°C , -6°C , -2°C , 1°C

from 1.07

- 26 The angles are co-interior. Kai says $a = 80$ when the other angle is 80° . Is that right? Explain.

- 27 The lines are parallel. Which angle, p or q , corresponds to the 110° angle?



from 2.07

Challenging

Questions 28–34

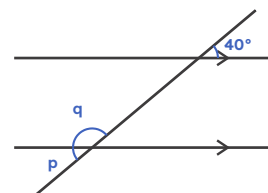
- 28 Alternate angles on parallel lines are $(3x + 15)^\circ$ and 60° . Find x .

- 29 The angles are co-interior. Mia says $a = 90$ when the other angle is 90° . Is that right? Explain.

- 30 Which is larger: -9 or -6 ?

from 1.03

- 31 Find p and q . Give a reason for each.



- 32 Find the supplement of 49° .

from 2.04

- 33 m and 159° are co-interior angles on parallel lines. Find m .

- 34 Alternate angles on parallel lines are $(3x + 15)^\circ$ and 111° . Find x .

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Two lines are parallel if corresponding angles are equal, or alternate angles are equal, or co-interior angles add to 180° .

Watch your teacher.
Copy every step.

E1 Alternate angles are 64° and 64° . Is $AB \parallel CD$?

E2 Co-interior angles are 105° and 85° . Is $AB \parallel CD$?

E3 Co-interior angles are $2x^\circ$ and 120° . Find x so that $AB \parallel CD$.

Easy

Questions 1–9

1 Co-interior angles are 110° and 70° . Is $AB \parallel CD$?

2 For $AB \parallel CD$, what must be true of corresponding angles?

3 Corresponding angles are 65° and 65° . Is $AB \parallel CD$?

4 For $AB \parallel CD$, what must be true of alternate angles?

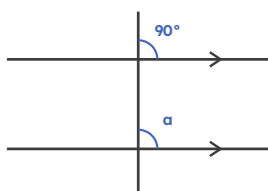
5 Alternate angles are 80° and 80° . Is $AB \parallel CD$?

6 For $AB \parallel CD$, what must be true of co-interior angles?

7 Alternate angles are 140° and 140° . Is $AB \parallel CD$?

8 Co-interior angles are 80° and 100° . Is $AB \parallel CD$?

9 The lines are parallel. Find α .



from 2.10

Easy

Questions 10–17

10 Alternate angles are 45° and 45° . Is $AB \parallel CD$?

11 Corresponding angles are 60° and 60° . Is $AB \parallel CD$? Give a reason.

____ = ____

yes: _____

12 Alternate angles are 85° and 85° . Is $AB \parallel CD$?

13 Which is larger: -16 or -5 ?

from 1.03

14 Alternate angles are 100° and 100° . Is $AB \parallel CD$? Give a reason.

15 Corresponding angles are 135° and 135° . Is $AB \parallel CD$?

16 Co-interior angles are 125° and 55° . Is $AB \parallel CD$? Give a reason.

17 The lines are parallel. What is the co-interior angle to 36° ?

from 2.09

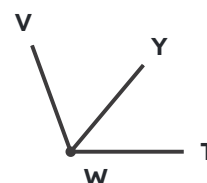
Medium

Questions 18–22

18 Corresponding angles are 30° and 30° . Is $AB \parallel CD$? Give a reason.

19 Co-interior angles are $2x^\circ$ and 134° . Find x so that $AB \parallel CD$.

20 Name the three angles at W.



from 2.01

21 Alternate angles are 30° and 30° . Is $AB \parallel CD$? Give a reason.

22 Co-interior angles are $2x^\circ$ and 130° . Find x so that $AB \parallel CD$.

2.11 Proving lines are parallel (continued)

YEAR 7 · CHAPTER 2: ANGLES · PAGE 2 OF 2

Medium

Questions 23–31

- 23 Co-interior angles are $2x^\circ$ and 148° . Find x so that $AB \parallel CD$.

- 24 Corresponding angles are 75° and 75° . Is $AB \parallel CD$? Give a reason.

- 25 What is the opposite of 28?

from 1.01

- 26 Alternate angles are $(3x + 10)^\circ$ and 88° . Find x so that $AB \parallel CD$.

- 27 To draw a reflex angle of 210° , what angle do you draw first?

from 2.02

- 28 Co-interior angles are $2x^\circ$ and 122° . Find x so that $AB \parallel CD$.

- 29 Alternate angles are 80° and 85° . Is $AB \parallel CD$? Give a reason.

- 30 Complete the rule: alternate angles on parallel lines ...

from 2.10

- 31 Alternate angles are $(3x + 10)^\circ$ and 34° . Find x so that $AB \parallel CD$.

Challenging

Questions 32–39

- 32 Alternate angles are $(3x + 10)^\circ$ and 58° . Find x so that $AB \parallel CD$.

- 33 A line crosses AB and CD . One angle at AB is 70° . The co-interior angle at CD is 115° . Are the lines parallel? Explain.

- 34 $-12 - (6 - 1)$

from 1.08

- 35 Priya says $AB \parallel CD$ because co-interior angles are both 125° . Is that right? Explain.

- 36 What letter shape do alternate angles make?

from 2.08

- 37 Alternate angles are $(3x + 10)^\circ$ and 85° . Find x so that $AB \parallel CD$.

- 38 A line crosses AB and CD . One angle at AB is 80° . The co-interior angle at CD is 105° . Are the lines parallel? Explain.

- 39 Find the angle between the clock hands at 5:00. Classify it.

from 2.03

Challenging

Questions 40–47

- 40 A line crosses AB and CD . One angle at AB is 60° . The co-interior angle at CD is 125° . Are the lines parallel? Explain.

- 41 Priya says $AB \parallel CD$ because co-interior angles are both 115° . Is that right? Explain.

- 42 Start at -1 . Move 3 right. Where do you land?

from 1.02

- 43 Alternate angles are $(3x + 10)^\circ$ and 40° . Find x so that $AB \parallel CD$.

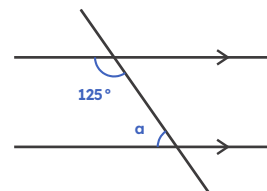
- 44 Find the supplement of 17° .

from 2.04

- 45 Co-interior angles are $2x^\circ$ and 146° . Find x so that $AB \parallel CD$.

- 46 A line crosses AB and CD . One angle at AB is 55° . The co-interior angle at CD is 130° . Are the lines parallel? Explain.

- 47 The lines are parallel. Find α .



from 2.09